

## Original article

## A deep learning framework for photovoltaic power forecasting in multiple interconnected countries

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## ABSTRACT

Accurate photovoltaic (PV) power forecasting is essential for enabling utilities, grid operators, and market participants to anticipate solar generation patterns and adjust operational strategies. Recently, Deep Learning (DL) has emerged as a promising approach for improving PV forecasts. However, training DL models typically require large datasets, which are often limited in PV power forecasting, restricting their full potential. Incorporating additional data from multiple heterogeneous sources, such as different countries, can help address this limitation but – at the same time – introduces challenges related to distribution shifts that can negatively impact forecasting accuracy. The rapid expansion of PV capacity further complicates forecasting, as generation patterns shift with new installations. Existing approaches often face challenges in effectively utilizing data from multiple sources. The main novelty of this paper lies in proposing a DL-based PV forecasting framework that integrates both foundational and country-specific training. This approach enables the framework to leverage data from multiple interconnected countries, improving generalization and enhancing forecasting accuracy. Additionally, we introduce an adaptation mechanism that dynamically recalibrates the forecasting model in response to changes in PV capacity, without the need for retraining, allowing our framework to seamlessly handle such variations. The proposed framework, evaluated on data from three interconnected countries (Greece, Bulgaria, and Romania), demonstrates significant improvements in accuracy by exploiting cross-country data and adapting to evolving conditions. Using data from these three interconnected countries is particularly important since each country's domestic PV production can directly affect the clearing prices and the energy mix of the neighboring countries. This work highlights a critical yet underexplored research direction, demonstrating the potential for significantly more accurate DL-based PV forecasting by effectively utilizing big data, in line with recent advances in DL across various fields.

## Introduction

Climate change nowadays constitutes a critical challenge that significantly affects global markets, regulations, policies, and investments. Renewable energy sources (RES) play a pivotal role in addressing the impacts of climate change due to their sustainable nature and potential to mitigate carbon emissions [1]. In recent years, significant steps have been taken towards the integration of RES into energy markets, and as a result, the energy landscape has been transformed with RES occupying a prominent place in the energy mix [2]. Photovoltaic (PV) power, derived from solar energy, has witnessed remarkable growth and adoption globally, driven by advancements in technology and supportive policies and according to European Commission constitutes the fastest-growing energy source [3]. In

interconnected power networks, where electricity flows across regions and national borders, accurate PV generation forecasting becomes even more critical. Having reliable forecasting models can support grid operators in managing real-time power flows more effectively, reducing reliance on expensive reserves and mitigating the risk of congestion in transmission lines. By improving short-term and day-ahead forecasts, TSOs can better coordinate cross-border electricity exchanges and optimize the dispatch of flexible resources such as battery energy storage systems (BESS) [4] and demand response. Furthermore, for policy-makers, enhanced forecast accuracy facilitates the development of robust regulatory frameworks and investment strategies that account for the variability and uncertainty of solar generation. This is particularly important in achieving high penetration of renewables while maintaining system reliability and market stability in interconnected grid

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environments, minimizing negative effects, such as the cannibalization effect [5].

A plethora of methodologies has been proposed in order to provide the most accurate PV forecasts, with deep learning (DL) proving more powerful than the alternatives [7]. Ren et al. [8] proposed a quad-kernel deep convolutional neural network model to enhance intra-hour PV power forecasting accuracy. Wang et al. (Jianzhou) [9] presented a comprehensive predictive system using automatic optimization of variational mode decomposition and multi-objective optimization algorithms. Pang et al. [10] combined artificial neural network (ANNs) and recurrent neural network (RNN) models to assess DL algorithms for solar radiation prediction. Kuo et al. [11] introduced a new approach using sky images to determine cloud cover for short-term PV power forecasting. An image analysis algorithm is developed based on on-site whole sky imager (WSI) data.

Zhang et al. [12] investigated DL hybrid technologies for ultra-short-term PV power forecasting. This includes feature engineering, DL-based point prediction, and error correction modules. Visser et al. [13] evaluated 12 diverse models predicting day-ahead power production aligned with market conditions, ranging from regression to DL and physical-based methods. Nguyen Trong et al. [14] proposed a novel approach for short-term PV power prediction combining Transformer Neural Network with CNNs, incorporating Variational Mode Decomposition (VMD) for data preprocessing to improve accuracy. Akhter et al. [15] investigated power output forecasting for photovoltaic systems using a hybrid deep learning approach compared to other models. By analyzing real data over several years, the study demonstrates that the proposed model outperforms alternative methods in accuracy and robustness for various PV technologies. Scott et al. [16] compared various machine learning algorithms to predict the performance of a PV system. Qu et al. [17] introduced a new hybrid model using GRUs to forecast distributed PV power generation. The approach leverages local historical data to predict one-day-ahead solar power at five-minute intervals. Qin et al. [18] proposed a novel approach combining ground and satellite data via DL to improve solar PV output predictions. Zhang et al. [19] introduced a novel deep reinforcement learning framework incorporating outlier detection via CatBoost and a twin adaptive convolutional deterministic policy gradient (TACDPG) model for accurate, interpretable PV power prediction, achieving reduced error and superior performance. Matera et al. [20] developed and validated a network of ANNs to predict hourly global PV power output using temperature and solar radiation data.

Perera et al. [21] examined two novel deep-learning-based methods for regional forecasting, incorporating both aggregated and individual time series along with weather data. Khan et al. [22] proposed a novel stacked ensemble algorithm, which combined ANN and LSTM models to forecast solar energy. Brester et al. [23] employed an ANN trained on weather observations and tested on numerical weather prediction (NWP) data to simulate operational PV power forecasting. Wang et al. (Kejun) [24] addressed the challenge of accurately forecasting photovoltaic power due to the inherent variability of solar energy. It explores various models, including convolutional neural networks, long short-term memory networks, and a hybrid model combining both.

Sarmas et al. [25] suggested a *meta-learning* approach to enhance one-hour-ahead deterministic forecasts for PV systems. It dynamically combined forecasts from various DL models to identify optimal conditions for each model. Sadeghi et al. [26] explored LSTM and GSA-LSTM methods for predicting PV power output across various timeframes. Abou Houran et al. [27] proposed a composite model that blends LSTM and Swarm Intelligence (SI) optimization techniques to accurately predict short-term PV output, overcoming limitations of traditional methods. Luo et al. [28] introduced an adaptive LSTM (AD-LSTM) framework with a two-phase learning strategy and sliding window algorithm to improve day-ahead photovoltaic power generation forecasting accuracy and dynamically address concept drift using real-time data. Liu et al. [29] developed a novel hybrid model using modal reconstruction and ensemble empirical mode decomposition is proposed

for short-term photovoltaic power forecasting.

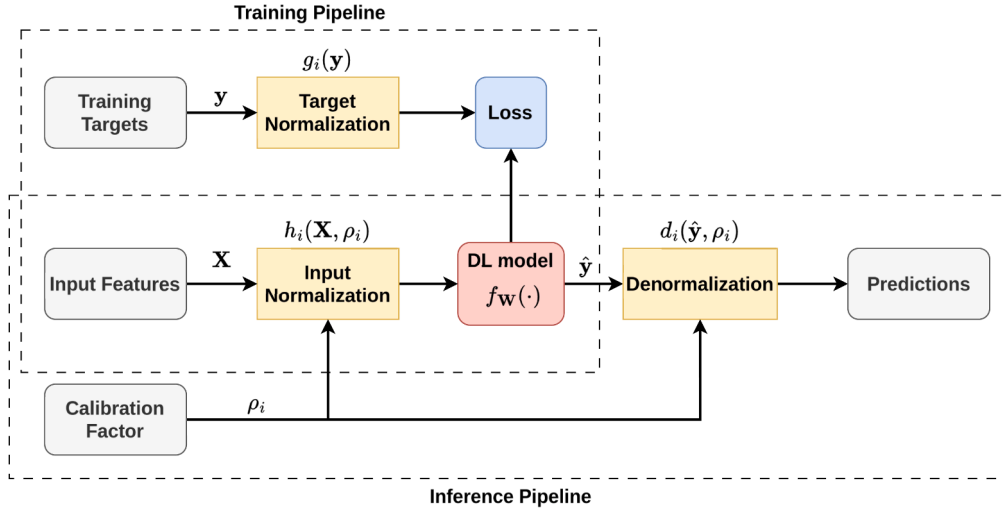
Zhang et al. [30] introduced an advanced prediction system combining decomposition algorithms and DL for ultra-short-term wind and photovoltaic power forecasting. The authors propose an enhanced decomposition method using fuzzy entropy and particle swarm optimization (PSO) to automatically adjust parameters for diverse energy data characteristics. Mirza et al. [31] presented an innovative hybrid model that combines and ResNet architectures with a Modified Recurrent Neural Network (MRNN). Chen et al. [32] proposed a novel approach to regional wind-photovoltaic combined power generation forecasting using a multi-task learning framework. The framework employs Maximum Information Coefficient (MIC) to identify key meteorological features and analyzes the complementarity and correlation between wind and PV power generation.

While deep learning (DL)-based approaches have shown strong performance in PV power forecasting, a common limitation is their dependency on large, high-quality datasets to achieve optimal results [33]. This poses a significant barrier for day-level, country-scale forecasting tasks, where the amount of historical data is inherently limited, particularly in smaller or newly solar-active countries. Moreover, existing approaches typically focus on single-country data, overlooking the potential benefits of multi-country learning despite the fact that countries with similar solar profiles could offer valuable shared information. However, directly applying models trained on data from one country to another often fails due to distribution shift, where differing environmental, technological, and temporal factors reduce transferability and forecast accuracy. Additionally, the rapid and continuous increase in installed PV capacity, driven by the global transition to renewable energy, creates non-stationarity in the data that renders static models obsolete over time. Current methods generally require full re-training when the data distribution changes, which is resource-intensive and impractical for real-time operations. These limitations highlight a critical research gap: the need for forecasting approaches that can leverage cross-country data while accounting for distribution shift, and adapt to changes in PV deployment without full model retraining.

This work proposes a novel DL framework to derive accurate PV power generation forecasting for three interconnected countries and in particular Greece, Bulgaria and Romania. While many previous works have developed DL models to forecast PV generation on a national level, to the best of authors knowledge it is the first time that such an analysis is carried out at a regional level. This is particularly important since these three countries are interconnected and each one's PV profile directly affects the electricity imports/exports among them and therefore whether the markets will be coupled or there will be distinct national clearing prices. Furthermore, the proposed approach could be a critical tool for the TSOs to gain comprehensive insights on the expecting PV power generation and based on that to proceed in the calculation of important parameters, such as the expected activation of baseload power plants and the net electricity transfer capacity between the interconnected countries. At the same time, the proposed framework incorporates an adaptation mechanism that enables end-users to easily adjust and re-calibrate the forecasting models when the installed PV capacity is increased without the need for time consuming and costly operations, such as re-training the forecasting model.

Considering the above context, the main contributions of this work are fourfold:

- i. We propose a novel DL-based PV power forecasting framework that can exploit data from multiple interconnected countries to improve forecasting accuracy. The proposed framework is model agnostic, i.e., it can be used with any DL model for forecasting PV power.
- ii. The proposed framework can tackle distribution shift phenomena caused by using data from different countries by decomposing training into two distinct phases, allowing for both exploiting all the available data, as well as training country-specific models.



**Fig. 1.** The proposed pipeline involves a) an input normalization module (employed in both training and inference), b) a target normalization module (employed only during training), and c) denormalization module (employed only during inference). Note that the proposed pipeline also employs a calibration factor for the normalization process, which allows for directly using the trained DL model even when distribution shift phenomena occur (e.g., when the installed PV capacity is increased).

- iii. We propose a novel adaptation mechanism to re-calibrate existing models to new conditions arising from expanding PV capacity in different countries.
- iv. We extensively evaluate the proposed framework on three interconnected countries, demonstrating a) the improvements that can be obtained when exploiting additional training data, as well as b) the ability of the proposed method to adapt to new situations without re-training.

The rest of this article is structured as follows: Section 2 provides an analytical presentation of the proposed methodology and the adopted solution strategy. Section 3 indicates the applicability of the proposed approach and evaluates the results obtained. Section 4 summarizes the main concluding remarks and proposes potential future directions.

## Methodology

In this section, we present the proposed DL-based forecasting framework, introducing the notation and problem definition (Section 2.1), presenting the developed architecture (Section 2.2) and deriving a training algorithm (Section 2.3).

### Notation and problem definition

Let  $y_t^{PV}$  denote the PV power injected into the grid of a specific country at hour  $t$ . We want to predict the 24-hour PV power vector for the day ahead defined as  $\mathbf{y}_{t+T_F}^{PV} = (y_{t+T_F}^{PV}, y_{t+T_F+1}^{PV}, \dots, y_{t+T_F+23}^{PV})^T \in \mathbb{R}_+^{24}$ , where  $T_F$  denotes the prediction horizon, i.e., how far into the future we want to make a prediction. Apart from relying on PV power variables in a solely autoregressive manner, DL models usually also rely on additional variables to increase the accuracy of predictions. Typical examples of such variables are weather conditions [34], weather forecasts [35], and other RES generation [32]. In Section 3 we evaluate the impact of using different combinations of such variables under the proposed framework. Therefore, the forecasting problem tackled in this paper is defined as:

**Definition 1. (PV power forecasting)** Given a window of  $T_H$  hours that contain the most recent measurements of  $N_F$  variables correlated with PV power generation,  $\mathbf{X}_t = (\mathbf{x}_{t-T_H+1}; \dots; \mathbf{x}_{t-1}; \mathbf{x}_t)^T \in \mathbb{R}^{T_H \times N_F}$ , where  $\mathbf{x}_t \in \mathbb{R}^{N_F}$  denotes the measurements of the  $N_F$  input variable at time  $t$  (e.g.,

weather, PV power, etc.), predict the  $T_F$  – hour ahead PV power vector  $\mathbf{y}_{t+T_F}^{PV}$ .

In this work, we aim to train a DL model, denoted as  $\hat{\mathbf{y}}_{t+T_F}^{PV} = f_W(\mathbf{X}_t)$ , where  $\mathbf{W}$  denotes the trainable parameters of the model, in order to estimate the PV power vector  $\mathbf{y}_{t+T_F}^{PV}$ . The level of correlation between the input features  $\mathbf{X}_t$  and the prediction target  $\mathbf{y}_{t+T_F}^{PV}$ , the amount of historical data used to fit the forecasting model  $f_W(\mathbf{X}_t)$ , along with its expressive power (i.e., capacity to model complex non-linear phenomena) and ability to withstand overfitting phenomena [36] determine to a significant degree the accuracy of the predictions.

The model  $f_W(\mathbf{X}_t)$  is fitted by minimizing a divergence metric over a training set  $\mathcal{X}_{train}$  of  $N$  samples that consist of pairs of input measurements  $\mathbf{X}_t$  and the corresponding ground truth predictions  $\mathbf{y}_{t+T_F}^{PV}$ :

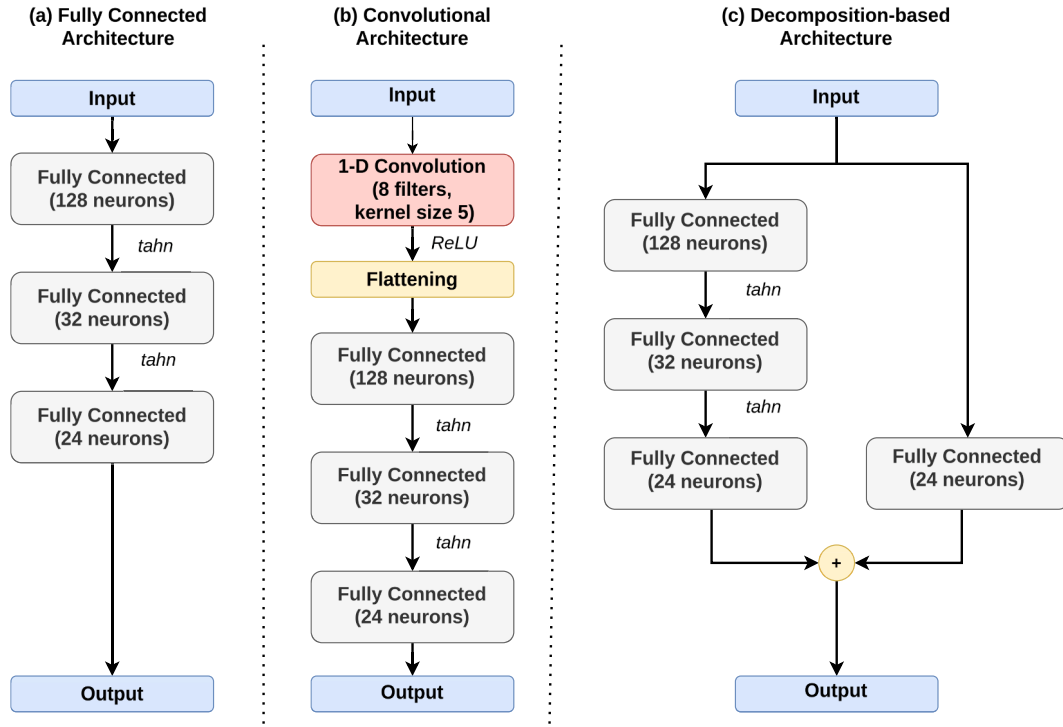
$$\min_{\mathbf{W}} \frac{1}{N} \sum_{\mathbf{X}_t, \mathbf{y}_{t+T_F}^{PV} \in \mathcal{X}_{train}} L(f_W(\mathbf{X}_t), \mathbf{y}_{t+T_F}^{PV}).$$

In this work, we employ the mean squared error (MSE) loss for fitting the model,  $L(\hat{\mathbf{y}}_{t+T_F}^{PV}, \mathbf{y}_{t+T_F}^{PV}) = \|\hat{\mathbf{y}}_{t+T_F}^{PV} - \mathbf{y}_{t+T_F}^{PV}\|_2^2$ , where  $\|\mathbf{a}\|_2$  denotes the  $l^2$  norm of a vector  $\mathbf{a}$ . In contrast with the rest of the literature, in this work we tackle the case of having data from multiple interconnected countries. Therefore, we denote as  $\mathcal{X}_{train}^i$  the training data that are collected from the  $i$ -th country, while we use the notation  $\mathbf{X}_{i,t}$  and  $\mathbf{y}_{i,t+T_F}^{PV}$  to similarly refer to the input features and ground truth collected from the  $i$ -th country.

### Modelling approach

The proposed architecture is outlined in Fig. 1. First, the input data are fed into a normalization module, which normalizes the input data based both on each country's statistics and re-calibration parameters. Then, a series of DL-based layers follows to analyze the input data and make appropriate predictions. Finally, the predictions are projected into the final prediction space using a denormalization module based on the applied normalization scheme. Applying appropriate normalization schemes is critical to ensure that the employed DL models will generalize efficiently and avoid chasing moving targets [37].

The normalization process is even more critical in the case in which data from multiple countries is used. In such a case, the data should be projected into a common representation space [38], to allow the model



**Fig. 2.** The proposed framework is model agnostic. In this paper we consider three different types of architectures: a) a simple fully connected architecture, b) a convolutional architecture, and c) a decomposition-based architecture.

to generalize the knowledge captured from one country into another country. In this work, we apply min–max normalization which restricts the data into the range 0 to 1. Data from different countries are separately normalized. Note that this choice is very intuitive since it enables to directly relate PV generation between different countries. Indeed, a value of “1” indicates the maximum production that can be achieved for any given country, while “0” indicates zero production. In this way, the model can readily generalize the captured knowledge between different countries without any additional mechanism. Therefore, the normalized data is calculated as  $\hat{\mathcal{X}}_{train}^i = \{ (h_i(\mathbf{X}_{i,t}), g_i(\mathbf{y}_{i,t+T_f}^{PV})) \mid (\mathbf{X}_{i,t}, \mathbf{y}_{i,t+T_f}^{PV}) \in \mathcal{X}_{train}^i \}$ , where the functions  $h_i(\cdot)$  and  $g_i(\cdot)$  are responsible for normalizing the input and targets for the  $i$ -th country. More specifically,  $h_i(\mathbf{X})$  is defined as

$$h_i(\mathbf{X}) = \frac{\mathbf{X} - \mathbf{x}_{i, \min}}{\mathbf{x}_{i, \max} - \mathbf{x}_{i, \min}},$$

where  $\mathbf{x}_{i, \min} \in \mathbb{R}^{N_f}$  and  $\mathbf{x}_{i, \max} \in \mathbb{R}^{N_f}$  denote the vectors containing the minimum and maximum values per feature for the  $i$ -th country and the operation of subtraction and division are applied elementwise after appropriately “broadcasting” the vectors into the same dimension as the corresponding matrices. The target normalization function is defined similarly as  $g_i(\mathbf{y}) = \frac{\mathbf{y} - \mathbf{y}_{i, \min}}{\mathbf{y}_{i, \max} - \mathbf{y}_{i, \min}}$ , where  $\mathbf{y}_{i, \min} \in \mathbb{R}^{24}$  and  $\mathbf{y}_{i, \max} \in \mathbb{R}^{24}$  denote the vectors containing the minimum and maximum values per feature for the  $i$ -th country. To project the predictions of the DL model back to the original space, we also employ a denormalization function  $d_i(\hat{\mathbf{y}})$  defined as:

$$d_i(\hat{\mathbf{y}}) = (\mathbf{y}_{i, \max} - \mathbf{y}_{i, \min}) \odot \hat{\mathbf{y}} + \mathbf{y}_{i, \min},$$

where  $\odot$  denotes the element-wise multiplication operation.

Even though this mechanism can effectively exploit data from multiple countries for training DL models, it suffers from a significant limitation. When the installed PV power capacity is increased, e.g., due to new investments in renewable energy, as witnessed in two of the

countries examined in this study (refer also to the case study presented in Section 3), the normalization scheme is no longer effective, pushing the employed DL model outside its working region. In practice, this leads to values greater than 1 to be fed into the DL model, forcing it to extrapolate into unknown regions, where it has never encountered data during the training process. To mitigate this phenomenon, we introduce a scalar re-calibration factor  $\rho_i$  which expresses the expected discrepancy between the maximum observed power in the training set and in the actual deployment. Therefore, the normalization function  $h_i(\mathbf{X})$  is split into two different cases: a)  $h_{i, PV}(\mathbf{X}_{solar})$  for normalizing PV power-related variables and  $h_{i, other}(\mathbf{X}_{other})$  for normalizing the rest of the variables  $h_i(\mathbf{X}, \rho_i) = [h_{i, PV}(\mathbf{X}_{solar}, \rho_i); h_{i, other}(\mathbf{X}_{other})]$ , where  $\mathbf{X} = [\mathbf{X}_{solar}; \mathbf{X}_{other}]$ . Then, these normalization functions are defined as:

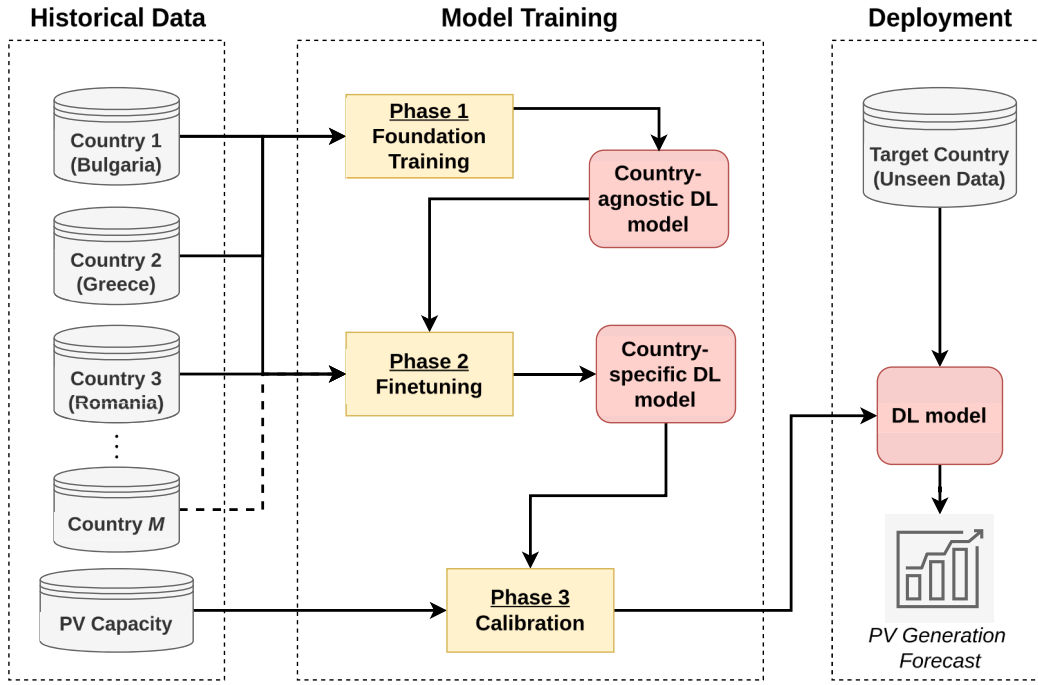
$$h_{i, solar}(\mathbf{X}_{solar}, \rho_i) = \frac{\mathbf{X}_{solar} - \mathbf{x}_{i, \min, solar}}{(\mathbf{x}_{i, \max, solar} - \mathbf{x}_{i, \min, solar})\rho_i^{-1}}, \text{ and}$$

$$h_{i, other}(\mathbf{X}_{other}) = \frac{\mathbf{X}_{other} - \mathbf{x}_{i, \min, other}}{(\mathbf{x}_{i, \max, other} - \mathbf{x}_{i, \min, other})},$$

where the minimum and maximum vectors are similarly defined as before. Then, the denormalization function is redefined as:

$$d_i(\hat{\mathbf{y}}, \rho_i) = \rho_i^{-1}(\mathbf{y}_{i, \max} - \mathbf{y}_{i, \min}) \odot \hat{\mathbf{y}} + \mathbf{y}_{i, \min}.$$

Selecting the value of  $\rho_i$  for each country during deployment is crucial to avoid the network operating in an extrapolation region. While, in principle, we can set this value directly based on the installed PV capacity, it is important to note that the relationship between installed capacity and output is not always linear. Additionally, although this value could be inferred using another model, fine-grained data required for this inference may not always be readily available. In addition to heuristically calibrating the value of  $\rho_i$  when PV expansions occur, we can also detect such events by observing the value of  $\mathbf{x}_{i, \max, solar}$ , which represents the maximum PV power generation. If this variable exceeds the corresponding value observed in the training data during inference, it suggests a potential need to increase the value of  $\rho_i$ . This adjustment



**Fig. 3.** The proposed training approach involves three stages: a) foundation training (generic modeling of PV generation), b) finetuning (adjusting models to specific countries), and c) calibration (adjusting the normalization process according to the expansion in PV capacity in specific countries).

can be made by setting  $\rho_i = \frac{x_{i, \max}^{\text{inference}}}{x_{i, \max}^{\text{solar}}}$ , where  $x_{i, \max}^{\text{inference}}$  refers to the maximum value observed during inference. Note that this is only an estimation, and further fine-tuning may be required in practice due to non-linear phenomena that can affect the calibration factor. This can be done by assessing the accuracy of the predictions and making necessary adjustments accordingly. Also, note that during deployment we only need to use the normalization function  $h_i(\mathbf{X}, \rho_i)$  and denormalization function  $d_i(\hat{\mathbf{y}}, \rho_i)$ . The normalization function  $g_i(\mathbf{y})$  is only employed during the training process and not needed during the deployment phase.

The proposed framework can work with any DL-based architecture. In this work, we employ three different architectures, as shown in Fig. 2. The first one is a fully connected one (Fig. 2a), composed of three fully connected layers. The first one is composed of 128 neurons, followed by a layer with 32 neurons and the final output layer with 24 neurons (one for each hour to be predicted). In the first two layers, we use *tanh* (hyperbolic tangent) as the activation function. The second architecture examined (Fig. 2b) is a convolutional one, composed of a 1-D convolutional layer with 8 filters (kernel size 5), followed by a flattening operation and three fully connected layers with 128 filters, 32 and 24 neurons. Finally, inspired by the recent decomposition-based models, such as D-Linear [39], we employ an architecture similar to the first one, but also using a linear projection layer directly connecting the input and output of the model, allowing for separately processing linear and non-linear components, as shown in Fig. 2c. Note that the data are appropriately flattened into vectors when fed into fully connected layers used as the inputs for the models.

### Training approach

All the DL models employed in this work are trained by employing stochastic gradient descent and employing the well-known Adam optimizer [40]. Instead of relying on data from one country, as most PV power forecasting methods do, in this work we employ data from multiple interconnected countries in order to better model the phenomena that govern and affect PV power generation. However, this is not

straightforward since different countries might exhibit different characteristics due to several reasons (e.g., the installed PV capacity is different between different countries, solar irradiance differs between different countries and local weather conditions can affect generation in different ways in different seasons). To account for such distribution shift phenomena that inevitably arise when using data from multiple countries, we first separately normalize the data from each individual country to align the distributions across countries (as detailed in Section 2.2) and then follow a transfer learning setup. This enables a DL model to initially learn the generic PV power generation patterns and then specialize in the phenomena that appear in specific countries.

To enable DL models to fully exploit the available data and achieve the goals mentioned before, we proposed a training process that consists of three distinct phases, as shown in Fig. 3. The first phase is called “Foundation Training” and aims to model the generic mechanism that governs PV generation irrespective of the country in which the model will be employed. This process uses data from all the available countries. The next phase is called “Finetuning” and aims to specialize the model to a specific set of countries. In this stage we continue training the previous model using data only from a specific. It is worth noting that during this stage might be beneficial to add data also from other countries as well, since this could have a positive regularization effect, reducing overfitting phenomena. Also, to avoid forgetting the previous modelled knowledge, a lower learning rate is used, following the standard transfer learning setting [41].

The model incorporates (country-related) geographical, climatic, and grid-specific factors during the country-specific fine-tuning process. In the foundation training phase, the model learns the underlying global dynamics affecting all countries. Then, during the country-specific finetuning, it adapts to the unique characteristics, including regional meteorological variations, of each country. Before deploying the model to a specific country, we employ recent data regarding any expansion of the installed PV power capacity (compared to the PV capacity that was installed when the training data were gathered) to re-calibrate the model (phase 3) using the proposed adaptation mechanism described in the previous section. Note that during calibration we do not re-train the DL model. Instead, we just estimate the optimal value for the scalar



**Table 1**

Comparing the forecasting performance of different DL models, as well as of the proposed approach, for three different countries. The proposed framework was applied to the best performing model for each country.

| Method   | RMSE               | MAE                | R <sup>2</sup>       |
|----------|--------------------|--------------------|----------------------|
| Bulgaria |                    |                    |                      |
| MLP      | 161.2 ± 0.8        | 127.4 ± 0.6        | 0.295 ± 0.003        |
| CNN      | 169.9 ± 2.3        | 133.3 ± 1.8        | 0.262 ± 0.008        |
| DMLP     | 130.9 ± 0.9        | 101.6 ± 0.9        | 0.404 ± 0.003        |
| Proposed | <b>114.9 ± 0.0</b> | <b>90.2 ± 0.0</b>  | <b>0.448 ± 0.000</b> |
| Greece   |                    |                    |                      |
| MLP      | 415.0 ± 1.0        | 331.7 ± 0.9        | 0.150 ± 0.002        |
| CNN      | 447.9 ± 3.3        | 357.8 ± 2.4        | 0.091 ± 0.006        |
| DMLP     | 314.3 ± 0.4        | 242.3 ± 0.3        | 0.301 ± 0.002        |
| Proposed | <b>286.0 ± 0.2</b> | <b>213.0 ± 0.2</b> | <b>0.358 ± 0.000</b> |
| Romania  |                    |                    |                      |
| MLP      | 50.2 ± 0.0         | 39.5 ± 0.0         | 0.347 ± 0.009        |
| CNN      | 49.2 ± 0.2         | 38.7 ± 0.1         | 0.349 ± 0.013        |
| DMLP     | 49.3 ± 0.1         | 38.4 ± 0.0         | 0.215 ± 0.011        |
| Proposed | <b>48.6 ± 0.1</b>  | <b>38.3 ± 0.1</b>  | <b>0.367 ± 0.011</b> |

calibration factor  $\rho_i$  for each country.

### Case study

In this Section we present a case study on three interconnected countries, namely Greece, Bulgaria and Romania. First, we present the collected data and evaluation setup in Section 3.1. Then, we proceed (Section 3.2) by presenting the quantitative evaluation of the proposed framework, while qualitative results are presented and discussed in Section 3.3. We also provide an additional ablation study and a sensitivity analysis for the proposed method in the supplementary material.

#### Data, setup, and evaluation metrics

We collected a dataset from three interconnected European countries (Greece, Bulgaria and Romania), while the data collection period ranged from 1 Jan 2020 to 29 Dec 2023 using data provided by Transmission System Operators (IPTO, ESO, Transelectrica) and Energy Exchanges (HENEx, IBEX, OPCOM), as well as European Centre for Medium-Range Weather Forecasts (ECMWF). The dataset consists of hourly measurements of a) PV generation, b) wind generation, c) average country-wide temperature and d) total grid load. Any missing values were filled using linear interpolation. The first three years were used to train all the models, while the last year was used to evaluate their performance on unseen data. The input window was set to  $T_H = 24$ . After data cleaning to ensure that data within each month are continuous in all cases, this split led to a total of 922 training instances and 306 test samples. We employed the min-max normalization process detailed in Section 2.2 for all the experiments conducted to ensure a fair comparison between different approaches.

All the models were trained for 1000 epochs using batch size of 256 samples, unless otherwise stated. We followed the same setup for both the proposed method and the baseline approaches to ensure a fair comparison, i.e., we first trained each model for 500 epochs with a learning rate of 0.001 and then 500 additional finetuning epochs followed using a reduced learning rate of 0.0001.

The prediction horizon was set according to the hourly electricity day-ahead (DA) market auctions (at 12:00 the day before the delivery day). We also trained models to perform forecasting for the intra-day market (at 15:00 and 22:00 the day before the delivery day, as well as at 10:00 within the delivery day). For the last intra-day prediction, we only evaluated the model on the future predicted generation (last 12 h). These correspond to critical time points where stakeholders make various decisions.

For all the experiments we report the average evaluation metrics over these four prediction horizons unless we explicitly state otherwise.

**Table 2**

Ablation study evaluating the impact of using data from multiple interconnected countries for training DL models for PV power forecasting, as well as the impact of individual features. The R<sup>2</sup> metric is reported. All models were trained for 200 epochs.

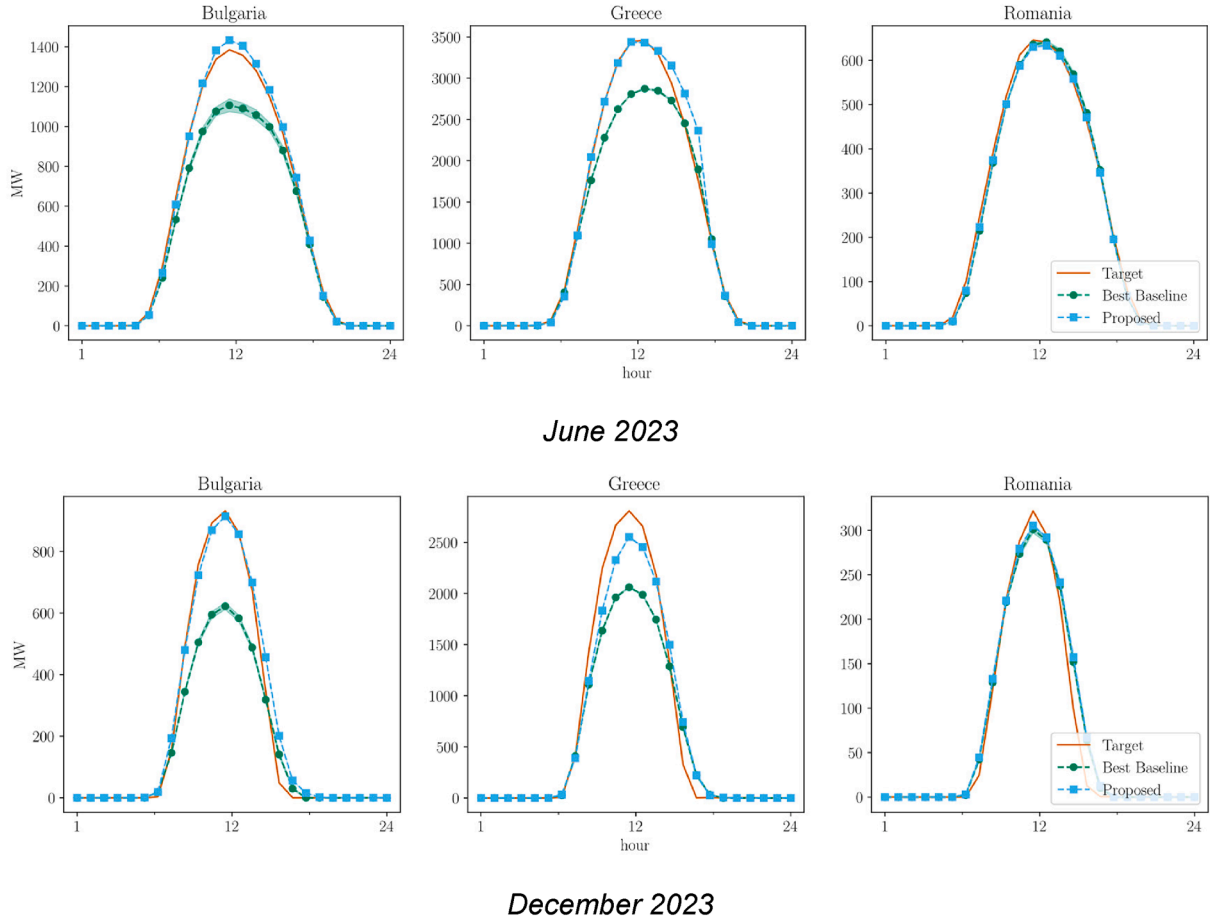
| Features                | Single Country | Multi Country (Proposed) |
|-------------------------|----------------|--------------------------|
| Bulgaria                |                |                          |
| PV                      | 0.305 ± 0.027  | <b>0.373 ± 0.001</b>     |
| PV + Wind               | 0.347 ± 0.026  | <b>0.377 ± 0.002</b>     |
| PV + Temperature        | 0.285 ± 0.019  | <b>0.330 ± 0.000</b>     |
| PV + Load               | 0.327 ± 0.030  | <b>0.369 ± 0.002</b>     |
| PV + Load + Temperature | 0.303 ± 0.010  | <b>0.327 ± 0.001</b>     |
| PV + Wind + Temperature | 0.317 ± 0.004  | <b>0.335 ± 0.001</b>     |
| PV + Wind + Load        | 0.342 ± 0.011  | <b>0.375 ± 0.001</b>     |
| Greece                  |                |                          |
| PV                      | 0.216 ± 0.004  | <b>0.296 ± 0.001</b>     |
| PV + Wind               | 0.243 ± 0.006  | <b>0.309 ± 0.000</b>     |
| PV + Temperature        | 0.142 ± 0.004  | <b>0.286 ± 0.001</b>     |
| PV + Load               | 0.201 ± 0.003  | <b>0.292 ± 0.001</b>     |
| PV + Load + Temperature | 0.145 ± 0.003  | <b>0.284 ± 0.001</b>     |
| PV + Wind + Temperature | 0.167 ± 0.003  | <b>0.303 ± 0.001</b>     |
| PV + Wind + Load        | 0.224 ± 0.002  | <b>0.307 ± 0.001</b>     |
| Romania                 |                |                          |
| PV                      | 0.122 ± 0.044  | <b>0.167 ± 0.009</b>     |
| PV + Wind               | 0.173 ± 0.008  | <b>0.175 ± 0.003</b>     |
| PV + Temperature        | 0.160 ± 0.010  | <b>0.188 ± 0.031</b>     |
| PV + Load               | 0.135 ± 0.008  | <b>0.151 ± 0.012</b>     |
| PV + Load + Temperature | 0.184 ± 0.034  | <b>0.184 ± 0.021</b>     |
| PV + Wind + Temperature | 0.188 ± 0.038  | <b>0.209 ± 0.030</b>     |
| PV + Wind + Load        | 0.161 ± 0.018  | <b>0.165 ± 0.015</b>     |

Each experiment was repeated 10 times, and the average and standard deviation are reported, unless otherwise stated. We use three commonly used evaluation metrics, i.e., Root Mean Square Error ("RMSE"), Mean Average Error ("MAE"), and Coefficient of Determination ("R<sup>2</sup>"). For the RMSE and MAE smaller values indicate better performance, while for the R<sup>2</sup> coefficient larger values indicate better performance. These metrics offer a comprehensive evaluation of both overall accuracy and robustness in PV power forecasting. The reader is referred to the supplementary material for details on the definition of these metrics and their interpretation in the context of PV forecasting.

#### Quantitative evaluation

First, in Table 1, we report the forecasting performance of four individual models, including the three backbone architectures introduced in Section 2.2. The fully connected architecture is abbreviated as "MLP," the convolutional architecture as "CNN," and the decomposition-based architecture inspired by the state-of-the-art D-Linear approach as "DMLP." These variants represent a broad spectrum of deep learning architectures. Based on the evaluations, we selected the best backbone architecture for each country to which the proposed framework was applied. At this point, it is worth noting that the proposed framework could potentially be applied to traditional machine learning approaches, such as Random Forests. However, due to the superior performance of deep learning approaches in recent years, we focus our experimental evaluation on these latter approaches.

Several interesting conclusions can be drawn from the results reported in Table 1. First, the DMLP approach is a strong performer, confirming the findings of other studies on the forecasting abilities of decomposition-based approaches [39], outperforming all the backbone architectures in two countries (Bulgaria and Greece). In Romania CNNs seem to lead to slightly better performance. Using the proposed approach on the other hand significantly outperforms all the evaluated models under all evaluation metrics. RMSE is reduced from 130 to 115 (about 10 % improvement) in the case of Bulgaria and from 314 to 286 (about 9 % improvement) in the case of Greece. Smaller improvements are observed in the case of Romania (RMSE is reduced from 49.3 to



**Fig. 4.** Comparing the average PV power forecasts for the three countries for June 2023. The proposed method is compared to the best performing baseline for each country. The first prediction horizon (DA) forecast is reported.

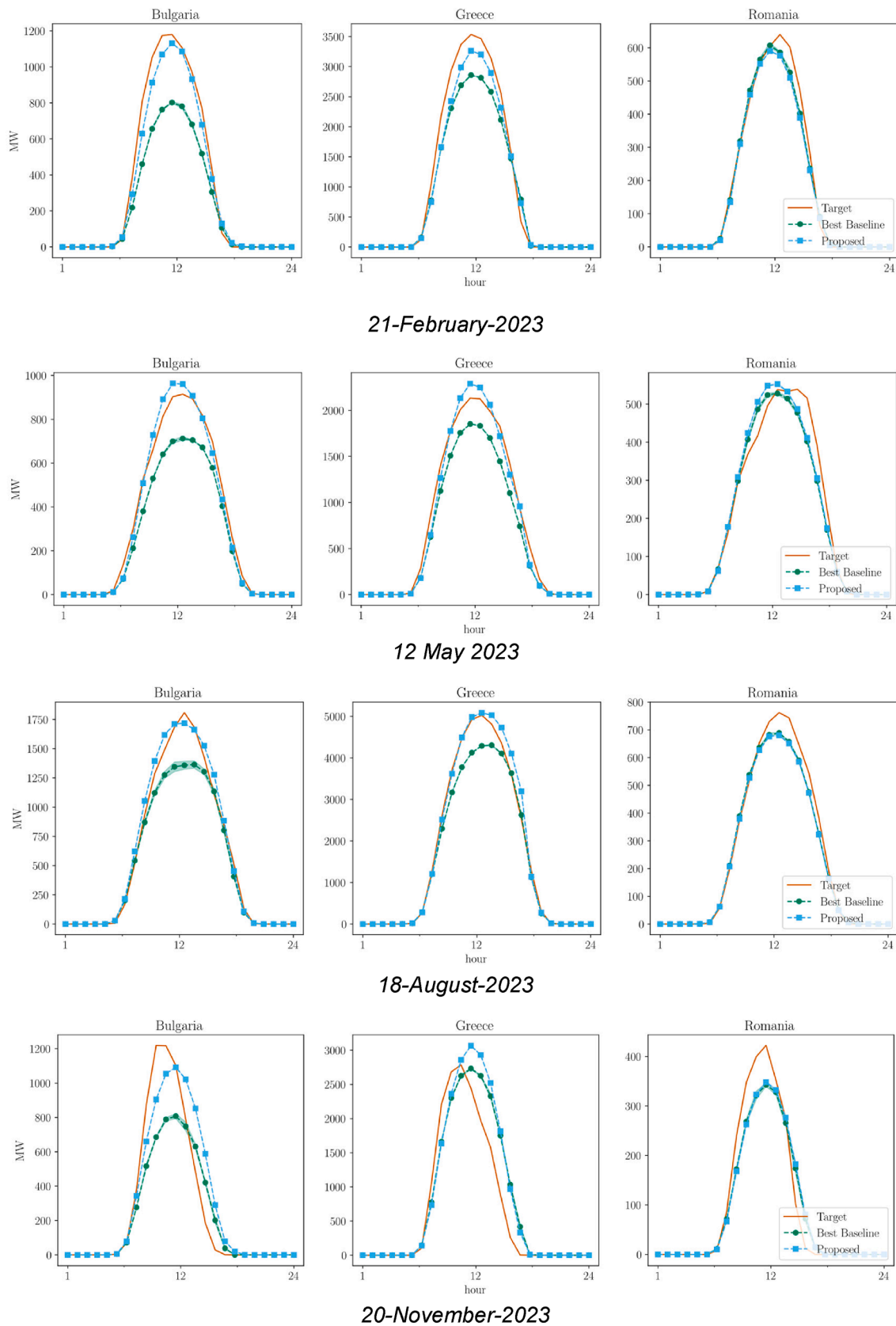
48.6). We conjecture that the smaller improvements observed in the case of Romania are due to the different conditions that exist in Romania compared to Bulgaria and Greece. More specifically, Romania has had a much smaller increase in installed PV capacity in recent years (e.g., compared to Bulgaria which almost doubled the installed capacity in 2023 [42]), as well as different weather patterns. Therefore, even though in the cases of Greece and Bulgaria we observe significant improvements by exploiting data from multiple countries, the improvements are smaller in the case of Romania. We believe that collecting data from countries with similar patterns to those observed in Romania has the potential to lead to similar improvements also in Romania.

To further confirm that the observed improvements are due to the use of additional training data instead of just employing the proposed adaptation mechanisms, we also performed a series of ablation experiments where we a) removed the adaptation mechanisms completely and b) employed different sets of input features to evaluate their impact in the performance of the models. The evaluation results of this ablation studies are reported in Table 2. We evaluated seven different combinations of input features. In all the evaluated cases employing additional data from multiple interconnected countries leads to improved forecasting accuracy. Also, note that different combinations of features lead to different forecasting performance. Combining PV and wind features seems to perform the best in our setup, possibly exploiting interrelations between the generation of different RES. Again, Romania seems to require different handling compared to Bulgaria and Greece, with the combination of PV power, wind power and average country-wide temperature features leading to the best forecasting performance. Therefore, for the results reported in Table 1 we employed the best combination of features for each individual country.

An additional ablation study provided in supplementary material reveals interesting dynamics regarding the generalization of the proposed approach under different fine-tuning strategies. Specifically, for Romania and Bulgaria, performing additional country-specific finetuning after foundation training countries yields the best overall performance. However, for Greece, the model trained directly with multi-country data, i.e., data from Greece, Romania, and Bulgaria, performs slightly better. A possible explanation is that Romania and Bulgaria exhibit unique dynamics that necessitate finetuning after foundation training, whereas Greece demonstrates more consistent behavior throughout the year. In this case, additional finetuning may lead to overfitting rather than improving model performance.

#### Qualitative evaluation

In Fig. 4 we provide the average PV power production during June and December 2023, along with the forecast provided by the best performing baseline architecture (as reported in Table 1) and the proposed method. In Romania all methods perform similarly, with the baseline and the proposed method closely matching the actual production. On the other hand, we observe that in Bulgaria and Greece the baseline significantly underestimates the production, both in summer and winter. This phenomenon can be attributed to the additional installed PV capacity in those countries in contrast with Romania. The proposed method, equipped with an adaptation mechanism, can easily adjust to such conditions, and closely match the actual production in the average case. For example, the forecast provided by the proposed method for peak power in Greece (3464 MW) closely matches the actual power (3441 MW), while the next best-performing approach underestimated



**Fig. 5.** Comparing the average PV power forecasts for the three countries for different indicative days. The proposed method is compared to the best performing baseline for each country. The first prediction horizon (DA) forecast is reported.



the peak power (2874 MW).

Next, we proceed by examining indicative days across the year in Fig. 5. Similar observations can be made as before. Both models seem to forecast the PV production more easily for Romania. For Bulgaria and Greece again the proposed method better matches the actual PV production compared to the next best-performing approach. It is also worth noting that in absence of accurate weather forecasts, most methods fail to accurately predict fine intra-day variations, and usually predict the most probable typical generation for the specific day. At the same time, the generation in the previous day (which is among the input features to the model) seems to be a very strong predictor that biases the model, as also discussed in the [supplementary material](#), where we also demonstrate the ability of the proposed adaptation mechanism to better match the actual PV power.

## Conclusions

In this work, we presented a novel DL-based PV power forecasting framework that can exploit data from multiple interconnected countries to improve forecasting accuracy. To this end, we designed an appropriate training process enabling DL models to handle distribution shift phenomena that are caused when using data from different countries. The proposed framework is model agnostic, i.e., it can be used with any DL model for forecasting PV power, as demonstrated through experiments. To validate the proposed framework, we conducted extensive experiments in three interconnected countries, demonstrating the improvements that can be obtained when exploiting additional training data, as well as the ability of the proposed method to adapt to new situations without re-training. The obtained results highlight the importance of a well-known, yet less investigated area in DL-based PV power forecasting, i.e., designing approaches that can make the best use of the data, while also exploiting data from multiple potential heterogeneous sources. Recent spectacular advances in DL, such as foundation models [43], demonstrated that training models with a large amount of data is critical for achieving the best possible performance.

Despite the benefits of the proposed approach, there are also some limitations. First, the method requires access to temporally aligned PV power generation data along with corresponding weather conditions. While this requirement can increase training time, it remains reasonable, as demonstrated in the supplementary material. When operating under constrained settings, neural network quantization and distillation techniques can be employed to improve efficiency. Another key limitation is the need to uncover well-known PV generation dynamics in a purely data-driven manner. Integrating hybrid physics-based and AI-driven approaches could enhance data efficiency, enabling training with smaller datasets while also improving the interpretability of predictions. Finally, the dataset used in this study includes actual PV production, wind production, load, and temperature data, meaning that the effects of grid interconnections, including cross-border electricity flows and market coupling, are inherently captured. However, cross-border exchange data (e.g., import/export flows or transmission constraints) could be also incorporated into the model to explicitly model such effects.

## CRedit authorship contribution statement

**Christos N. Dimitriadis:** Writing – original draft, Validation, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Nikolaos Passalis:** Writing – original draft, Validation, Software, Methodology, Investigation, Formal analysis, Conceptualization. **Michael C. Georgiadis:** Writing – review & editing, Supervision, Funding acquisition, Conceptualization.

## Declaration of competing interest

The authors declare that they have no known competing financial

interests or personal relationships that could have appeared to influence the work reported in this paper.

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## Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.seta.2025.104330>.

## Data availability

Data will be made available on request.

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